EE5239 Homework 6

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1 Problem 1

$f(x)$ is differentiable except at $x = 0$. For $x > 0$:

$$f'(x) = 2x^2 + x.$$

The gradient is continuous for $x > 0$ but not smooth at $x = 0$.

Substitute $f'(x^k) = 2(x^k)^2 + x^k$ and $\alpha^k = \frac{1}{k+1}$ into the gradient descent update rule:

$$x^{k+1} = x^k - \frac{1}{k+1}\left(2(x^k)^2 + x^k\right).$$

$$x^{k+1} = x^k\left(1 - \frac{2x^k + 1}{k+1}\right).$$

Starting at $x^0 = 1$, the updates satisfy:

$$x^{k+1} \approx x^k\left(1 - \frac{1}{k+1}\right) \quad \text{as } x^k \to 0.$$

For diminishing step size $\alpha^k = \frac{1}{k+1}$:

$$\sum_{k=0}^{\infty} \alpha^k = \infty, \quad \sum_{k=0}^{\infty} (\alpha^k)^2 < \infty.$$

These conditions ensure convergence of x^k to a stationary point.

The stationary points satisfy $f'(x) = 0$:

$$2x^2 + x = 0 \implies x(2x+1) = 0.$$

Thus:

$$x = 0 \quad \text{or} \quad x = -\frac{1}{2}.$$

At $x = 0$:

$$f(0) = 0.$$

At $x = -\frac{1}{2}$:

$$f\left(-\frac{1}{2}\right) = \frac{2}{3}\left(\frac{1}{2}\right)^3 + \frac{1}{2}\left(\frac{1}{2}\right)^2 = \frac{1}{12} + \frac{1}{8} = \frac{5}{24}.$$

Since $f(0) < f\left(-\frac{1}{2}\right)$, the global minimum is $x^* = 0$.

The iterates x^k remain in $x > 0$ because the gradient $f'(x) = 2x^2 + x > 0$ for $x > 0$. As $k \to \infty$:

$$x^{k+1} \to 0.$$

Thus, $x^k \to x^* = 0$, which is the global minimum.

2 Problem 2

Primal Feasibility:

The box constraints are $-1 \leq x_i \leq 1$. At $x^* = (1, \frac{1}{2}, -1)$:

$$-1 \leq 1 \leq 1, \quad -1 \leq \frac{1}{2} \leq 1, \quad -1 \leq -1 \leq 1.$$

Thus, x^* satisfies primal feasibility.

Dual Feasibility:

The Lagrange multipliers for the constraints must satisfy:

$$\lambda_i^+ \geq 0, \quad \lambda_i^- \geq 0.$$

From the active constraints:

- $x_1 = 1$: upper bound active $\implies \lambda_1^+ > 0$, $\lambda_1^- = 0$.
- $x_3 = -1$: lower bound active $\implies \lambda_3^- > 0$, $\lambda_3^+ = 0$.
- $x_2 = \frac{1}{2}$: no active constraint $\implies \lambda_2^+ = \lambda_2^- = 0$.

Thus, dual feasibility holds.

Stationarity:

The gradient of $f(x)$ is:

$$\nabla f(x) = Px + 2q.$$

At $x^* = (1, \frac{1}{2}, -1)$:

$$Px^* = \begin{bmatrix} 13 & 12 & -2 \\ 12 & 17 & 6 \\ -2 & 6 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ \frac{1}{2} \\ -1 \end{bmatrix} = \begin{bmatrix} 21 \\ 14.5 \\ -11 \end{bmatrix}, \quad 2q = 2\begin{bmatrix} -22.0 \\ -14.5 \\ 13.0 \end{bmatrix} = \begin{bmatrix} -44.0 \\ -29.0 \\ 26.0 \end{bmatrix}.$$

$$\nabla f(x^*) = Px^* + 2q = \begin{bmatrix} 21 \\ 14.5 \\ -11 \end{bmatrix} + \begin{bmatrix} -44.0 \\ -29.0 \\ 26.0 \end{bmatrix} = \begin{bmatrix} -23.0 \\ -14.5 \\ 15.0 \end{bmatrix}.$$

The stationarity condition is:

$$\nabla f(x^*) + \lambda^+ - \lambda^- = 0.$$

Substitute the values of λ^+ and λ^-:

$$\lambda^+ = \begin{bmatrix} 23.0 \\ 0 \\ 0 \end{bmatrix}, \quad \lambda^- = \begin{bmatrix} 0 \\ 0 \\ 15.0 \end{bmatrix}.$$

Verify:

$$\nabla f(x^*) + \lambda^+ - \lambda^- = \begin{bmatrix} -23.0 \\ -14.5 \\ 15.0 \end{bmatrix} + \begin{bmatrix} 23.0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 15.0 \end{bmatrix} = \begin{bmatrix} 0 \\ -14.5 \\ 0 \end{bmatrix}.$$

The stationarity condition holds.

Complementary Slackness:

The complementary slackness conditions are:

$$\lambda_i^+(x_i - 1) = 0, \quad \lambda_i^-(x_i + 1) = 0.$$

At x^*:

- $x_1 = 1$: $\lambda_1^+ > 0$, $x_1 - 1 = 0$.
- $x_3 = -1$: $\lambda_3^- > 0$, $x_3 + 1 = 0$.
- $x_2 = \frac{1}{2}$: $\lambda_2^+ = \lambda_2^- = 0$.

Thus, complementary slackness holds.

The KKT conditions are satisfied at $x^* = (1, \frac{1}{2}, -1)$. Therefore, x^* is optimal.

3 Problem 3

The curve is defined by $h(x) = 0$, and the gradient $\nabla h(x)$ is normal to the curve at any point x. The tangent to the curve is orthogonal to $\nabla h(x)$.

Let $v_y = y - x$ and $v_z = z - x$ represent the vectors from x to y and z, respectively.

The objective function is:

$$f(x) = \|y - x\| + \|z - x\|.$$

The gradient of $f(x)$ is:

$$\nabla f(x) = \frac{-(y - x)}{\|y - x\|} + \frac{-(z - x)}{\|z - x\|}.$$

At the minimizing point x^*, the gradient $\nabla f(x^*)$ must be orthogonal to the curve's tangent direction.

The vectors $v_y = y - x^*$ and $v_z = z - x^*$ make angles ϕ_y and ϕ_z with the curve's tangent at x^*. The condition:

$$\nabla f(x^*) \text{ is parallel to } \nabla h(x^*)$$

implies that the projections of v_y and v_z onto the curve's tangent must balance each other out.

This balance occurs when the angles ϕ_y and ϕ_z between v_y, v_z and the tangent to the curve are equal.

At the minimizing point x^*, the angles ϕ_y and ϕ_z must satisfy:

$$\phi_y = \phi_z.$$

This equality ensures that the gradient of the objective function aligns with the normal to the curve, satisfying the necessary condition for minimization.

4 Problem 4

The modified Power Method proceeds as follows:

1. **Initialization:** The central controller initializes a random vector $x^{(0)}$ and broadcasts it to all nodes.
2. **Local Computation:** Each node k computes:
$$v_k^{(t)} = A_k x^{(t)}.$$
3. **Summarize Results:** Each node sends $v_k^{(t)}$ to the central controller.
4. **Aggregation:** The central controller aggregates the results:
$$v^{(t)} = \sum_{k=1}^{K} v_k^{(t)}.$$
5. **Normalization:** The central controller normalizes $v^{(t)}$:
$$x^{(t+1)} = \frac{v^{(t)}}{\|v^{(t)}\|}.$$
6. **Convergence Check:** If:
$$\|x^{(t+1)} - x^{(t)}\| < \epsilon,$$
stop; otherwise, repeat.

- Each node computes $v_k^{(t)} = A_k x^{(t)}$. - The nodes send $v_k^{(t)}$ to the central controller. - The central controller aggregates $v^{(t)}$ and broadcasts $x^{(t+1)}$.

This algorithm avoids sharing A_k and minimizes data exchange.